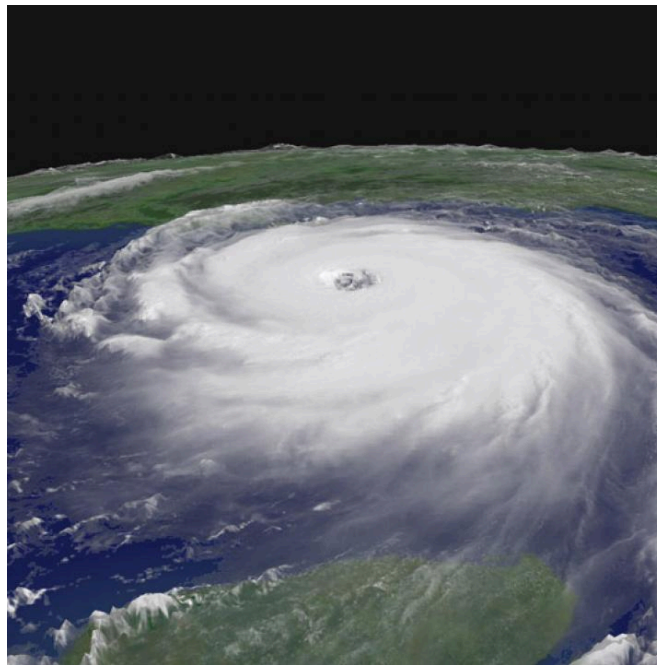


KTPeach

Hurricane Relief Application

Software Requirements Specification



CSCE 247: Software Engineering

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1. Introduction

Purpose

This project will develop a digital platform to help communities access and coordinate hurricane-relief resources. During and after a hurricane, information about shelters, food, medical care, supplies, and volunteer support is often scattered across many sources, making it difficult for people to quickly find the help they need. Our product will provide a centralized, easy-to-use place for affected residents to find reliable resources and for relief organizations or volunteers to better identify community needs. The goal is to improve communication, reduce confusion, and help support a faster, more organized recovery process.

2. Stakeholders

- Residents affected by hurricanes
- Local relief organizations
 - Shelters
 - Food banks
 - Nonprofits
- Volunteers
- Local government and emergency-management agencies
- KTPeach development team

Personas

Adam Watson

Age: 27

Occupation: Paramedic

Location: Miami, FL

Bio

Adam is a paramedic who specializes in natural disaster relief. He enjoys being able to help those in need, and constantly looks for opportunities to provide aid to everybody.

Goals

- Save more people
- Drive faster
- Get stronger

Frustrations

- Poor response time
- IV placement
- Traffic lights



Technology Savviness

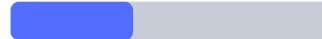
IT & Internet



Software



Mobile Apps



Social Media



Hobbies: Working out, hiking, and bird watching

Jonas Pork

Age: 52

Occupation: Small Business Owner

Location: Naples, FL

Bio

Jonas enjoys volunteering. He has lived through many hurricanes and has seen how it affects people and does as much as he can to help.

Goals

- Help people get out of hurricane trouble
- Locate nearby people that need assistance
- Deliver resources

Frustrations

- Outdated Info
- Cluttered Apps
- App fails under no signal

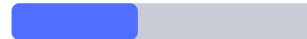


Technology Savviness

IT & Internet



Software



Mobile Apps



Social Media



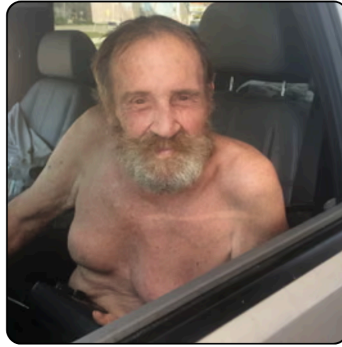
Hobbies: Working Out, Reading, Volunteering

Arthur Morgan

Age: 73

Occupation: Retired Officer

Location: Tampa, FL



Bio

Arthur underwent hurricane Helene and chose not to heed the warning due to living there for over 40 years and is now trapped with limited resources amidst the storm.

Goals

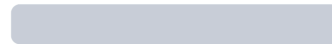
- Get Saved

Frustrations

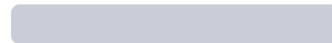
- Flooded Terrain
- No power
- Limited food/water

Technology Savviness

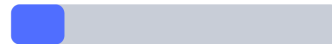
IT & Internet



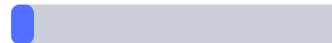
Software



Mobile Apps



Social Media



Hobbies: Golfing, Swimming, and Gardening

3. Constraints

Time Constraints

- This project must be completed within the course semester

Monetary Constraints

- This project has a development budget of \$0 and must use free tools, services, and APIs

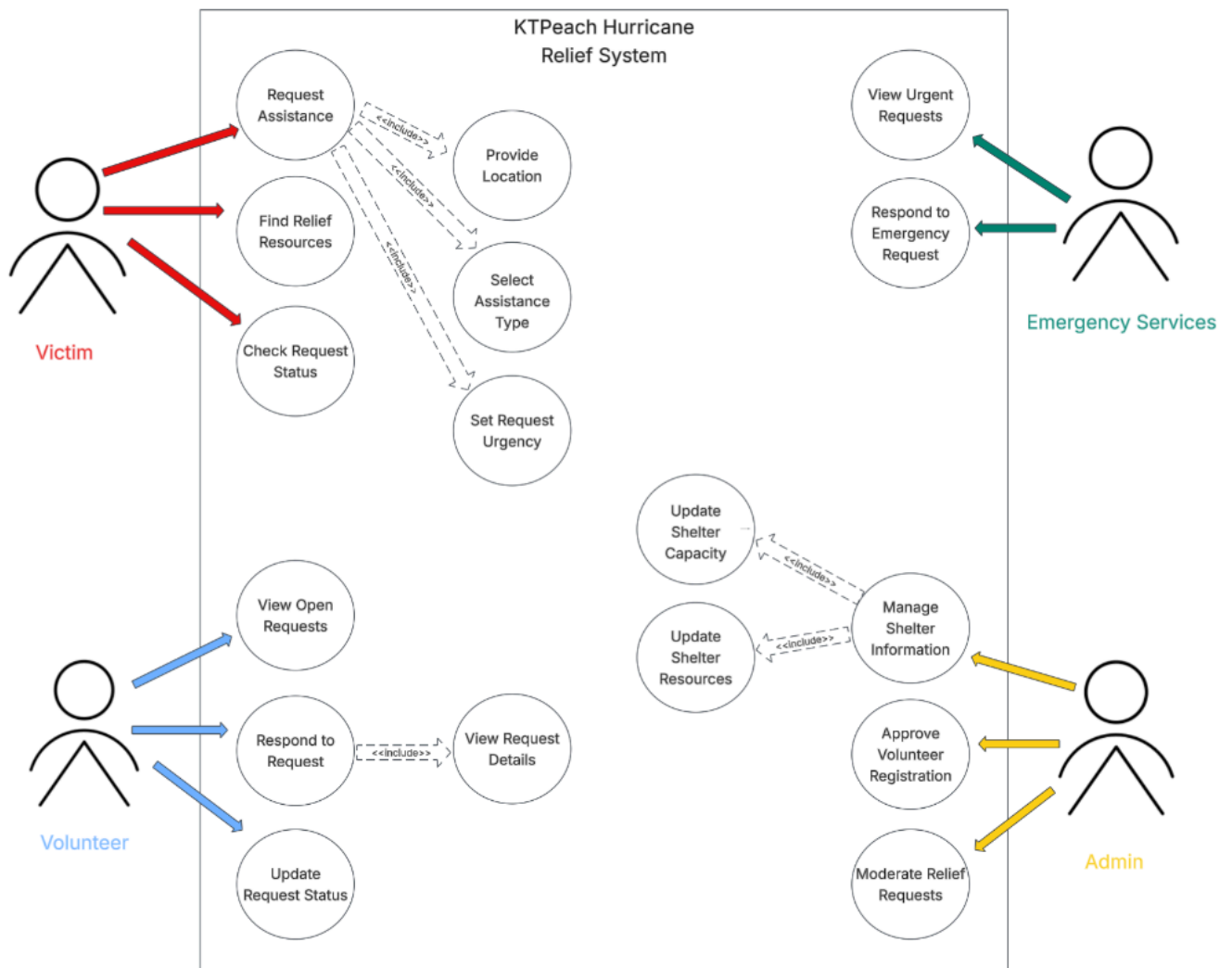
Technical Constraints

- The code must be done in Java and with the technology required by the course
- The platform must be accessible on desktop and mobile devices
- The project must protect user information and should not collect unnecessary personal data

4. Overall Description

This application must be accessible to affected residents, relief organizations, and volunteers across desktop and mobile devices. The app will collect and display centralized data on hurricane relief resources like emergency shelters, food distribution sites, medical aid, and volunteer opportunities. Users will be given options to search and filter available assistance based on their location, type of need and urgency level. The app will also manage assistance requests and track volunteer availability to ensure relief efforts are organized and distributed effectively during a storm.

5. Business Use Cases



6. Functional Requirements

Requirements Spreadsheet

7. Non-Functional Requirements

Look and Feel Requirements

- The system shall present high-priority relief requests with a clear visual indicator.
- The system shall display emergency information prominently.

Usability Requirements

- The system shall enable a first-time user to submit a relief request without prior training.
- The system shall provide text that is easy to understand during a stressful emergency situation.
- The system shall support users with visual impairments through accessible text contrast.

Performance Requirements

- The system shall display core information within three seconds under normal network conditions.
- The system shall remain responsive when multiple users access the system simultaneously.
- The system shall provide access to previously downloaded shelter information during an internet outage.

Maintainability and Support Requirements

- The system shall permit authorized shelter administrators to update shelter information without developer assistance.
- The system shall record the most recent update time for each shelter record.
- The system shall support current versions of Chrome, Safari, Edge, Firefox.

Security Requirements

- The system shall require authentication before a user accesses volunteer-only functions.
- The system shall restrict personal contact information to users involved in the related relief request.
- The system shall transmit personal data through encrypted connections.

Cultural Requirements

- The system shall provide Spanish translations for user-facing emergency instructions.
- The system shall use respectful language that avoids assumptions about a user's identity, family structure, or circumstances.

Legal Requirements

- The system shall comply with applicable federal, state, local privacy laws.
- The system shall display a disclaimer directing users to call 911 during an immediate life-threatening emergency

9. Competitive Analysis

	FEMA
Strengths:	Provides tutorials on how to create communication plans within a family, in addition to providing a packing list in case of an emergency. Also provides real-time weather and emergency alerts. Locates nearby shelters if evacuation is needed. Also tells you whether or not your location is viable for supplies and assistance. The app also follows federal guidelines for accessibility.
Weaknesses:	It's a mobile app and therefore requires a charged phone to use its features. Although the app has 24/7 response, data and message rates may apply and therefore isn't entirely cost-free. Prefers cash as opposed to any other payment method for donations. Heavily relies on trusted organizations which may limit the efficiency of volunteering. Also seems to be focused on civilians as opposed to having extensive features for EMS workers.
Target Audience:	Those in need, and civilians who want to volunteer/help/donate.

	e4e relief
Strengths:	Heavily focuses on financial assistance for teams and companies that lose labor during emergency situations. Supports many kinds of parameters when constructing a financial plan. Also stabilizes and keeps businesses afloat financially that may be unable to operate during emergency times.
Weaknesses:	Solely focuses on financial relief as opposed to any other kind of relief during an emergency. Can't be utilized unless you're an owner of a business/company.
Target Audience:	Companies and businesses in need of a financial support plan for their teams during lost hours in the event of a crisis.

	American Red Cross
Strengths:	Provides lots of helpful tips on how to stay safe before, during, and after any kind of emergency. Gives opportunities to those who want to volunteer for disaster relief. Also provides training programs and certifications to people on First Aid and CPR. Operates internationally as well, providing aid throughout the world.
Weaknesses:	Lots of things to look at on the website, and could possibly be hard to navigate for somebody who isn't the greatest at using the internet/technology.
Target Audience:	The general public and people who need help before/during/after an emergency situation.

Summary

	Strengths	Weaknesses	Audience
FEMA	• Real-time alerts • Preparation tips • Shelter locations	• Requires charged phone •	• Civilians • Trusted volunteer organizations
e4e relief	• Financial support	• Solely for businesses to create a financial plan • Provides one aspect of relief	• Business/Company owners
American Red Cross	• Preparation tips • Disaster relief team/volunteers • Safety training	• Expansive website with lots of features everywhere	• Civilians • Volunteers • EMS

The three products we analyzed each provide useful disaster-relief services, though their strengths serve different user needs. FEMA offers helpful features such as weather alerts, supply information, shelter locations, and accessibility resources. However, we identified a potential limitation: users may need a functioning mobile device to access certain features. Since hurricanes can cause power outages, our system should make critical information as accessible as possible for users with limited device access. e4e Relief provides strong financial-assistance support, but its narrower focus does not address the broader range of needs our system intends to support.

The American Red Cross provides a more comprehensive model for disaster relief by offering information and resources across many areas of recovery. Its website demonstrates the value of providing users with multiple forms of assistance in one location. However, the amount of information presented at once can feel overwhelming for users who need help quickly. Based on this analysis, KTPeach should provide a broad set of hurricane-relief resources while maintaining a simple, straightforward experience. The system should help users quickly locate relevant assistance without requiring them to navigate a complex or confusing interface.